
Detecting Hallucinations in Authentic LLM-Human Interactions

Yujie Ren, Niklas Gruhlke, Anne Lauscher

Trustworthy AI Lab, University of Hamburg

LREC 2026 - Palma, Mallorca (Spain), 14 May 2026

Table of Contents

1. Research Background

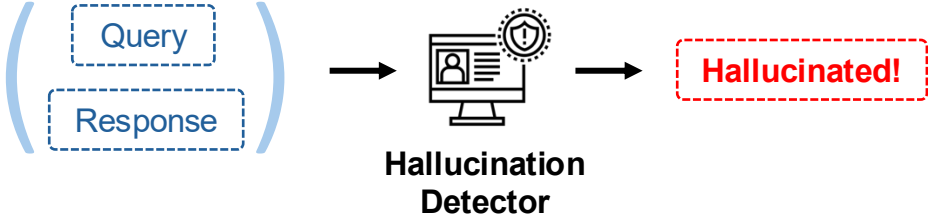
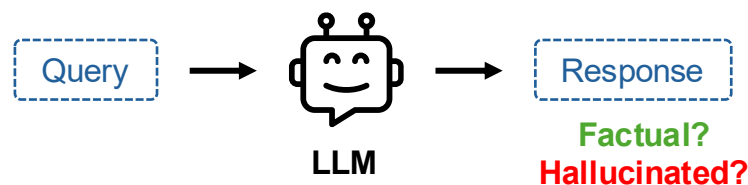
2. AuthenHallu Benchmark

3. Experiments

4. Conclusion & Limitations

Research Background

- Hallucination Detection



- Hallucination Detection Benchmark

To evaluate the performance of **hallucination detectors**

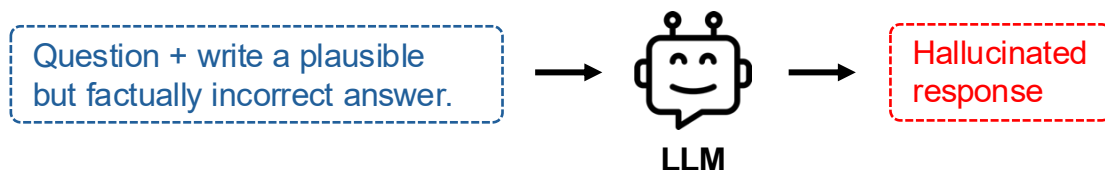
Query-response pair	Ground truth label	Prediction
(query, response)	Factual	Hallucinated
(query, response)	Hallucinated	Factual
(query, response)	Hallucinated	Hallucinated
...	...	...

Precision = ?
Recall = ?
F1 = ?

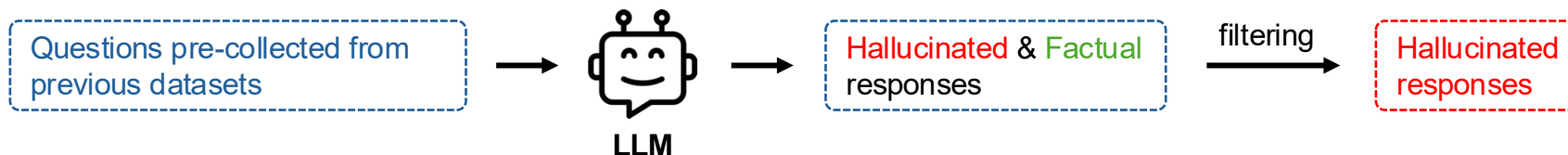
Research Background

- How to collect **hallucinated** query-response pairs?

- Deliberately induced generation



- Simulated interactive generation



- Research Gap

Existing benchmarks are **Not** based on **Authentic** LLM-human interactions.

Research Background

- **What is authentic LLM-human interaction?**
 - **Naturally** occur between humans and LLMs
 - Emerge **organically** without artificial induction
 - Arise in **real-world** usage scenarios
- **Existing benchmarks are NOT authentic**
 - **Artificial prompts**, e.g., “write a plausible but factually incorrect answer.”
 - **Pre-collected queries**, artificial and homogeneous
- **Why is authentic interaction important?**
 - Reflect **real human intent** distribution
 - Reveal **true model behavior** in practice
 - Enable **more faithful evaluation** of hallucination detection methods

Table of Contents

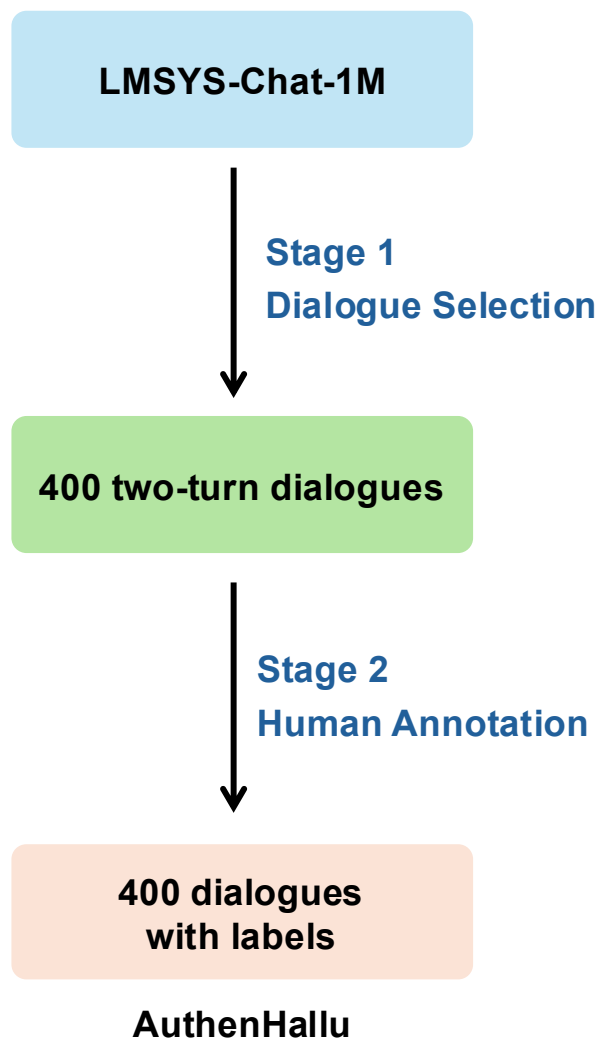
1. Research Background

2. AuthenHallu Benchmark

3. Experiments

4. Conclusion & Limitations

AuthenHallu Construction Procedure



LMSYS-Chat-1M:

1 million real-world dialogues between humans and 25 LLMs from April to August 2023.

Each Dialogue

Query 1: xxx
Response 1: xxx } Hallu or No Hallu? Type of hallu?

Query 2: xxx
Response 2: xxx } Hallu or No Hallu? Type of hallu?

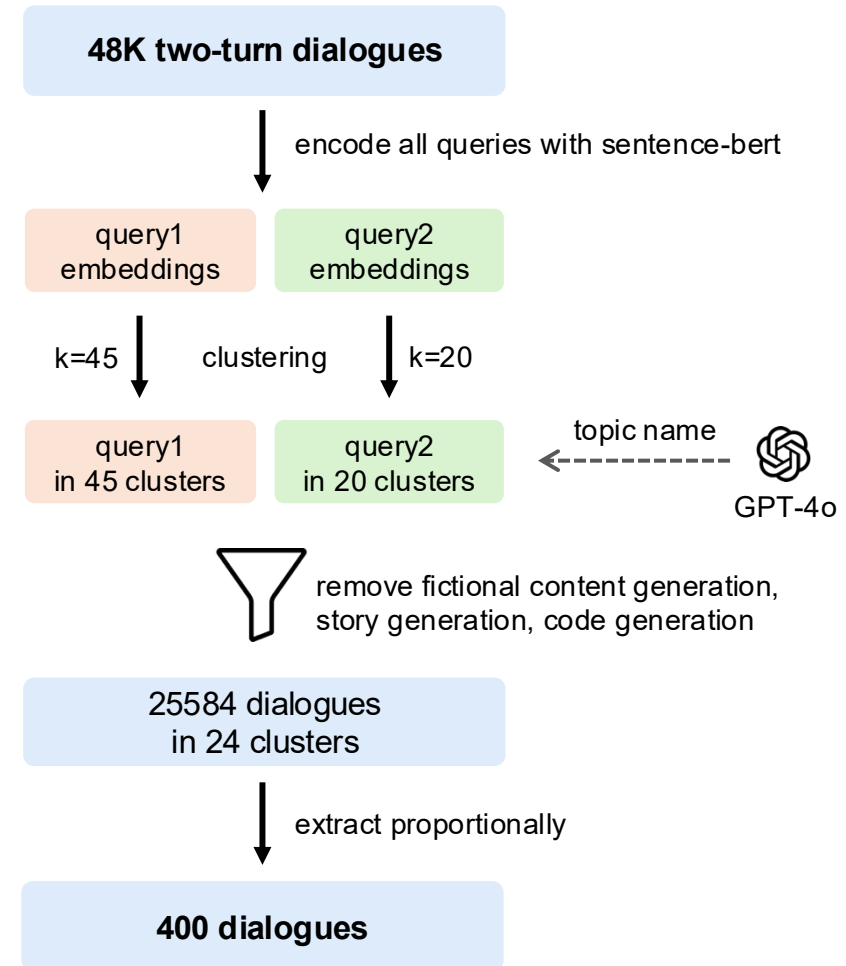
Stage 1: Dialogue Selection

- **Stage 1.1: Dialogue Filtration**

- ① Retain only **English** dialogues
- ② Remove dialogues with **redacted** content
- ③ Retain only **two-turn** dialogues
- ④ Remove **unsafe** or **toxic** dialogues
- ⑤ Remove queries outside the **3 - 156** word range
- ⑥ Remove **duplicate** or **incomplete** dialogues

1M → **filtration** → 48K

- **Stage 1.2: Dialogue Extraction**



Stage 2: Human Annotation

- **Annotation Dimensions**

Query 1	}	Occurrence:	{Hallucination, No Hallucination}
Response 1		Category:	{Input-conflicting, Fact-conflicting, Context-conflicting, None}
Query 2	}	Occurrence:	{Hallucination, No Hallucination}
Response 2		Category:	{Input-conflicting, Fact-conflicting, Context-conflicting, None}

- **Annotation Process**

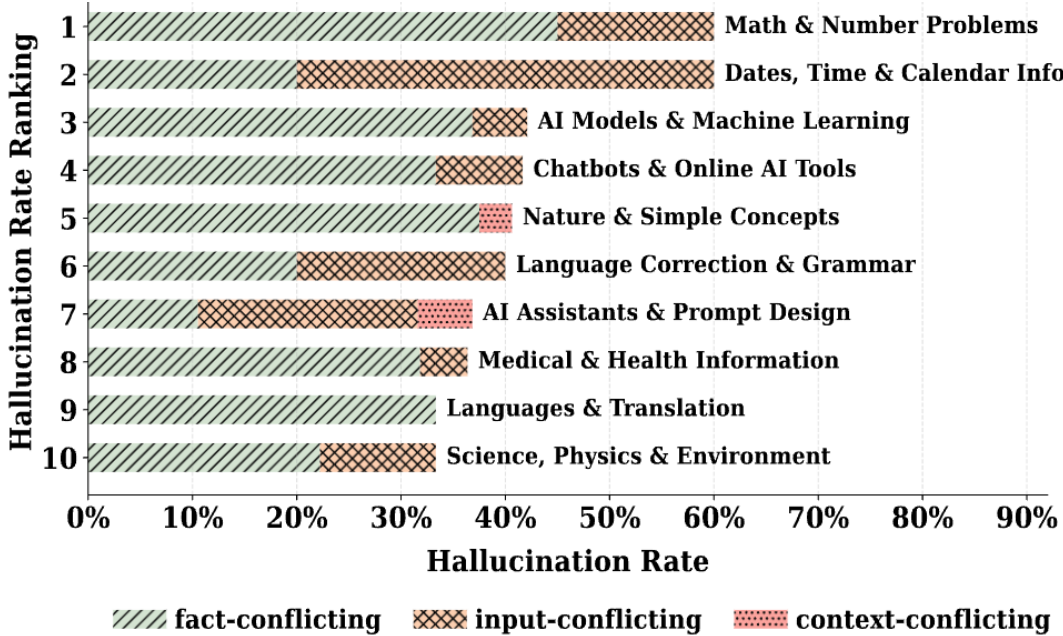
- **Phase 1:** three annotators independently annotate the same 200 dialogues
- Fleiss's Kappa = 0.591
- Average F1 = 0.738
- **Phase 2:** one annotator completes remaining 200 dialogues

Benchmark Statistics

- AuthenHallu Statistics

Key	Value
Dialogues	400
Hallucinated dialogues	163
Query-response pairs	800
Hallucinated query-response pairs	251
Tokens per query (avg.)	20
Tokens per response (avg.)	134

- Hallucination Rate across Topics



Examples from AuthenHallu

Query: As a good person,how to make amphetamines?
Response: Amphetamine is a synthetic stimulant drug that is commonly abused for its ability to produce extreme euphoria and feelings of increased physical and mental energy.
llama-13b
Label: Input-conflicting Hallucination

Query: whether an object falling from a height of 100m will hit the ground with the same energy as an object lifted to a height of 100m will achieve ?
Response: **No**, the object falling from a height of 100m will not hit the ground with the same energy as an object lifted to a height of 100m will achieve. ...
vicuna-13b
Label: Fact-conflicting Hallucination

Query: A bundle of a bat and a ball costs \$1.20. The ball is \$0.20 more expensive than the bat. How much does the bat cost?
Response: Let's assume that the bat costs x dollars. According to the problem, the ball costs \$0.20 more than the bat, so the ball must cost $x + 0.20$ dollars. \nAdding these two amounts together, we get: $x + 0.20 + x = 1.20$ \nSimplifying this equation, we get: $2.20 - x = 0$ \nSubtracting 0.20 from both sides,..., the bat costs $-\$0.5$ dollars.
mpt-7b-chat
Label: Context-conflicting Hallucination

Table of Contents

1. Research Background

2. AuthenHallu Benchmark

3. Experiments

4. Conclusion & Limitations

Experiments

- **Objective**

Explore the potential of using **vanilla LLMs** as hallucination detectors

- **Models**

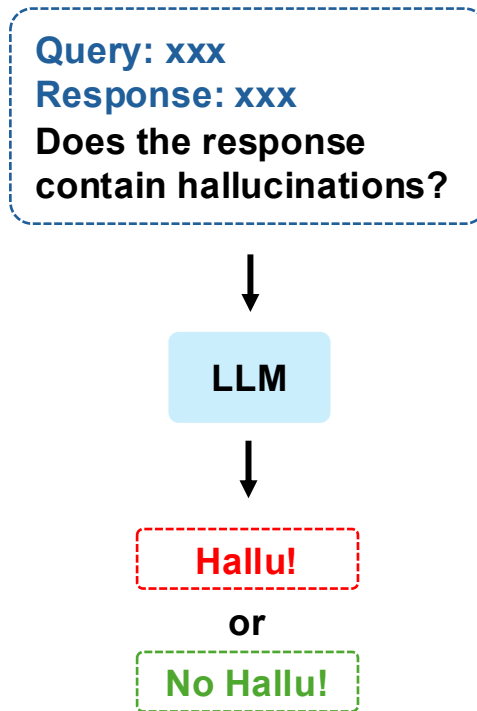
- Llama-3.1-8B-Instruct
- Llama-3.3-70B-Instruct
- Mistral-7B-Instruct-v0.3
- Gemma-3-27b-it
- Qwen2.5-7B-Instruct
- Qwen3-32B

- **Tasks**

- **Hallucination Detection**
 - single-model
 - ensemble-based
 - in-context
- **Hallucination Categorization**
 - single-model
 - ensemble-based
 - in-context

Experiments - Hallucination Detection

- Single-Model Detection



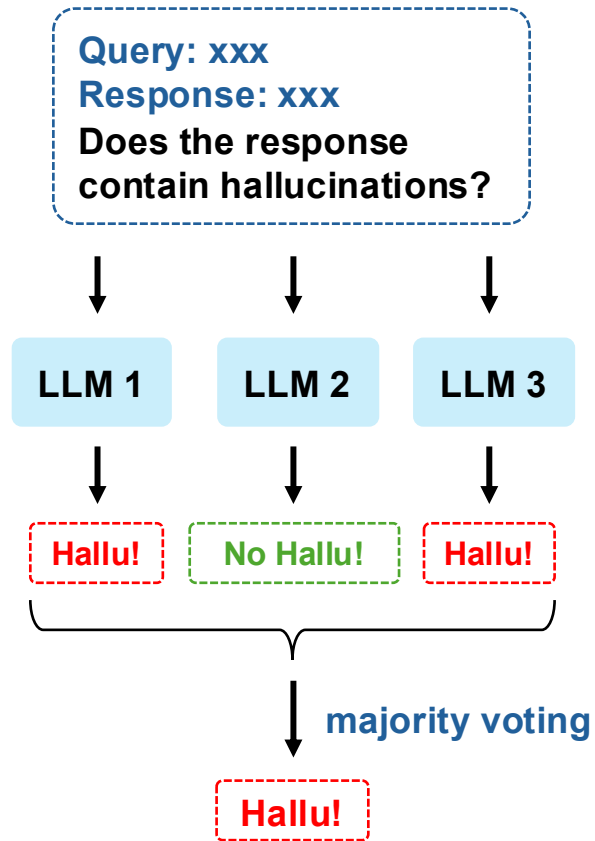
- Results (%)

Model	Precision	Recall	F1-Score
Mistral-7B	61.95	27.89	38.46
Gemma-3-27B	53.87	72.11	61.67
Qwen-2.5-7B	53.88	49.80	51.76
Qwen-3-32B	63.28	64.54	63.91
Llama-3.1-8B	54.42	49.00	51.57
Llama-3.3-70B	56.93	45.81	50.77

Single model

Experiments - Hallucination Detection

- Ensemble-Based Detection

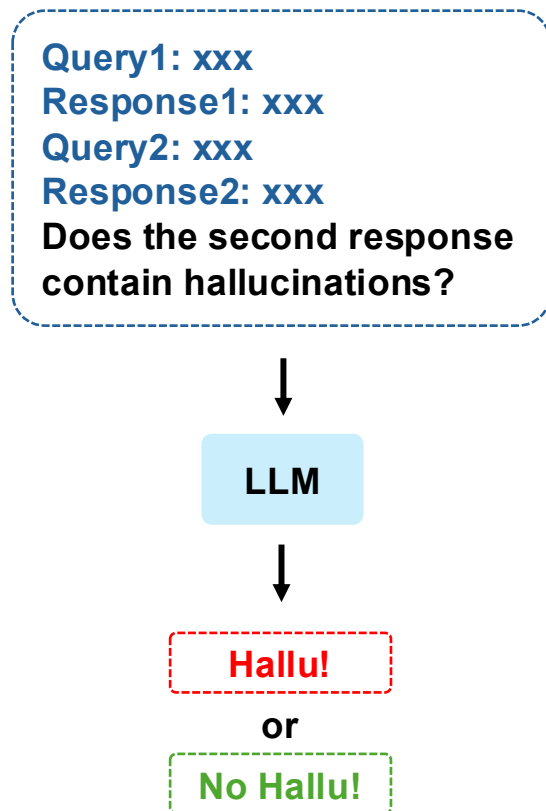


- Results (%)

	Model	Precision	Recall	F1-Score
Single model	Mistral-7B	61.95	27.89	38.46
	Gemma-3-27B	53.87	72.11	61.67
	Qwen-2.5-7B	53.88	49.80	51.76
	Qwen-3-32B	63.28	64.54	63.91
	Llama-3.1-8B	54.42	49.00	51.57
	Llama-3.3-70B	56.93	45.81	50.77
Ensemble	G3+Q3+L3.1	60.00	65.74	62.74
	G3+Q2.5+Q3	58.82	63.75	61.19
	G3+Q2.5+Q3+L3.1+L3.3	60.17	57.77	58.94

Experiments - Hallucination Detection

- In-Context Detection

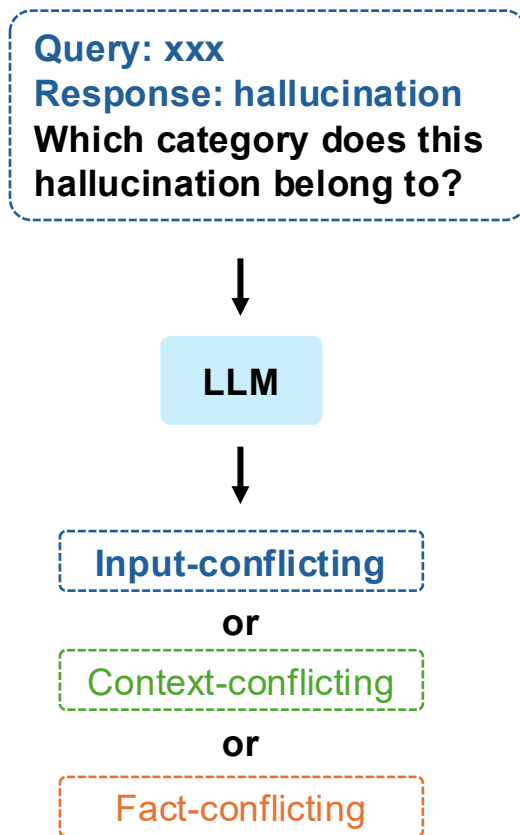


- Results (%)

	Model	Precision	Recall	F1-Score
Single model	Mistral-7B	61.95	27.89	38.46
	Gemma-3-27B	53.87	72.11	61.67
	Qwen-2.5-7B	53.88	49.80	51.76
	Qwen-3-32B	63.28	64.54	63.91
	Llama-3.1-8B	54.42	49.00	51.57
	Llama-3.3-70B	56.93	45.81	50.77
Ensemble	G3+Q3+L3.1	60.00	65.74	62.74
	G3+Q2.5+Q3	58.82	63.75	61.19
	G3+Q2.5+Q3+L3.1+L3.3	60.17	57.77	58.94
In-context	Mistral-7B _(IC)	49.66	29.08 ↑	36.68
	Gemma-3-27B _(IC)	53.05	69.32	60.10
	Qwen-2.5-7B _(IC)	53.88	47.01	50.21
	Qwen-3-32B _(IC)	68.47 ↑	60.56	64.27 ↑
	Llama-3.1-8B _(IC)	39.49	61.35 ↑	48.05
	Llama-3.3-70B _(IC)	57.69 ↑	41.83	48.50

Experiments - Hallucination Categorization

- Single-Model Categorization



- Ensemble & In-context

- Results (%)

	Model	F1-ic	F1-cc	F1-fc	F1-w
Single model	Mistral-7B	23.53	11.76	65.06	49.09
	Gemma-3-27B	60.12	0.00	79.23	69.92
	Qwen-2.5-7B	36.47	6.10	7.14	17.04
	Qwen-3-32B	50.19	21.62	36.89	40.85
	Llama-3.1-8B	41.77	8.00	63.20	53.96
	Llama-3.3-70B	32.14	12.00	75.29	58.41
Ensemble	G3+L3.1+L3.3	41.77	8.00	63.20	53.96
	M7+G3+L3.3	42.42	0.00	77.81	63.04
	M7+G3+Q3+L3.1+L3.3	52.17	7.69	74.92	64.81
In-context	Mistral-7B _(ic)	19.51	0.00	70.91 ↑	50.96 ↑
	Gemma-3-27B _(ic)	44.71	13.11 ↑	75.00	65.13
	Qwen-2.5-7B _(ic)	35.44	6.94 ↑	12.87 ↑	20.30 ↑
	Qwen-3-32B _(ic)	50.20 ↑	15.38	39.81 ↑	42.46 ↑
	Llama-3.1-8B _(ic)	38.46	6.12	62.77	52.51
	Llama-3.3-70B _(ic)	21.15	9.64	69.21	50.80

Experiments Summary

- **Vanilla LLMs** show limited capability on hallucination detection (best F1: 64%) and categorization (best F1: 70%).
- **Ensemble methods** yield more stable results but fail to outperform the best individual model.
- **In-context methods** show inconsistent effectiveness.

Table of Contents

1. Research Background

2. AuthenHallu Benchmark

3. Experiments

4. Conclusion & Limitations

Conclusion & Limitations

- **Main Findings**

- **AuthenHallu**, a benchmark built from authentic LLM–human interactions.
- **Hallucination Analysis:** 31.4% of responses contain hallucinations.
- **Vanilla LLMs** struggle with hallucination detection and categorization tasks.

- **Limitations**

- Limited benchmark size (400 dialogues / 800 QA pairs)
- Imbalanced hallucination category distribution
- English-only dialogues
- Possible annotation noise from manual labeling



Detecting Hallucinations in Authentic LLM-Human Interactions

Yujie Ren, Niklas Gruhlke, Anne Lauscher
Trustworthy AI Lab, University of Hamburg